

ODRE PQC

Product Overview & Security Architecture

ODRE PQC 0.3.1 RC1 · Release 63

Verified scope, authority, trust boundary, multi-platform delivery, FULL AUM and measured performance.

September 2026 · Release 63 technical baseline

1. Document authority and verdict

Release 63 is the Final Customer Delivery authority. The active Production pointer is Release 60; the two are intentionally distinct.

Authority	Verified state
Final Customer Delivery	Release 63 · g269 · 6/6 OS PASS
Active Production	Current 60 · Previous 59 · COMMITTED · Health PASS
Policy	Generation 1080 · Consumer Pin SYNCED
FULL AUM	g270 · A–R isolated PASS · destructive Production test NOT RUN BY POLICY

- Customer Delivery technical pass: YES; official supported OS: 6 of 6 PASS.
- Negative Gate: 18/18 PASS. Release 63, Sealed Core and g269 evidence remained byte-identical after closure.
- FULL AUM A–R was exercised with installed updater bytes and isolated fixtures; destructive Production injection was not run by policy.

2. Product purpose

ODRE PQC binds post-quantum protected execution to signed artifacts, runtime identity, policy generation, consumer pin, license, lease and unit authority. A failed prerequisite does not downgrade to plaintext or classical fallback.

- Data Plane: Commercial Native and the approved Sealed Core perform protected operations.
- Control Plane: policy, consumer pin, identity and commercial entitlement determine whether execution is authorized.
- Evidence Plane: manifests, receipts, journals, canonical status and reports preserve the basis for PASS, FAIL and NOT RUN.

3. Security architecture

Gateway, Native Runtime, Sealed Core, Policy, License and AUM are separate components but are joined by one authority at protected-request time.

- Missing or modified required artifacts fail before installation or apply.
- Policy and consumer pin mismatch produces zero protected execution and zero partial apply.
- Native/Core readiness or protected-request failure forbids commit.
- The Sealed Core is immutable, but it must be packaged, installed, loaded and exercised.

4. Protected request contract

An authenticated request is accepted only after entitlement, identity, policy synchronization, Native readiness and Core loading are confirmed.

- The business handler is invoked exactly once for each measured successful protected request.
- Unauthorized execution, plaintext fallback and implicit classical downgrade are zero.
- A health code alone is not sufficient; the protected request is part of the commit gate.

5. Release package trust chain

Selection is based on signed authority and exact SHA-256, never on the largest filename or most recent timestamp.

- Product archive → signed inventory/manifest → envelope/CMS → installer binding → installed-byte load → protected integration → final ZIP seal.
- Release 60/61/62 bytes were not overwritten. Release 63 is a monotonic successor that closes the Ubuntu 24.04 Native authority lineage.
- Installer execution, installation integration, runtime integration and Customer Delivery PASS are separate gates.

6. FULL AUM lifecycle

DISCOVER, VERIFY, STAGE, SWITCH, HEALTH, COMMIT, ROLLBACK and RECOVERY form a durable transaction.

- A single writer lock prevents concurrent apply transactions.
- Partially copied candidates are not promoted to VERIFIED.
- Atomic canonical status and a durable journal prevent torn JSON or a false COMMITTED state.
- Capacity is checked before staging; rollback assets are not deleted to make room.

7. Services and administration

Windows services remain under the SCM lifecycle. start_server does not bypass the service contract by launching executables or Python entrypoints directly.

- Already-running services are not restarted; only stopped services are started through SCM.
- Admin Update is authenticated, dynamic and read-only.
- Observed Production: current 60, previous 59, floor 60, policy 1080, pin SYNCED, COMMITTED/PASS, health 8000/18443 HTTP 200.

8. Official platform coverage

The same package, authority and protected-request contract passed on Windows Server 2022, Ubuntu 22.04, Ubuntu 24.04, Debian 12, Rocky Linux 9.8 and AlmaLinux 9.8.

OS	Result	Receipt SHA-256
Windows Server 2022	PASS	5E390E2062034FDE03BFA304D6A64EEAA191F79CB292E0EA913086CF5CB30489
Ubuntu 22.04	PASS	2E19C70F2992A1FA520E757DFC64A2FDB78DDEDBAB4E6AA4F943954D7B6E627D
Ubuntu 24.04	PASS	4AFBB78E6AB65D7D9389FA2B337C7E7350828AD69D827011CA399BAB226BC1B4
Debian 12	PASS	2597605241A07F53282EE0CEE6E5EF101919E178AAD00CBC06FF7FEBB2060293
Rocky Linux 9.8	PASS	701244BC819DA109AA190621F83EC91C36C5EDC0129041D80509F9AA952070AE
AlmaLinux 9.8	PASS	EBDBA21B870BD33032C26420669FC3F313890F45548B3369EB974C9E8FEB8563

- Windows-specific SCM/ACL and Linux service/permission adapters do not change the trust model.
- Containers, Kubernetes, other Windows versions and other Linux major versions are not claimed by this 6-OS verdict.

9. Fail-closed and negative testing

Tampered ZIP/SHA, CMS failure, inventory mismatch, policy drift, duplicate/lower generation, runtime failure and insufficient capacity must stop before unsafe success.

- REJECTED_PRE_APPLY means Production change is zero.
- Post-switch health failure triggers restoration of Product, policy, pins and runtime authority.
- Historical FAIL evidence is retained after the minimal fix and successful retest.

10. Evidence and interpretation

g269 proves the Final Customer Delivery closure; g270 proves the bounded FULL AUM and live read-only Admin contract.

- Installed updater regression: 208/208 PASS; drop-in/existing AUM: 14/14 PASS; 12-A/Admin: 19/19 PASS.
- Drive upload/readback and two Notion history append/readbacks passed. Drive re-download SHA was not run by policy because no integrity anomaly was observed.
- No FIPS 140-3, Common Criteria, universal performance or universal platform claim is made.

Measured performance evidence

The figures below are from historical g108 Windows Server 2022 loopback candidate testing. They were not re-measured on Final Release 63 and must not be generalized to WAN or customer hardware.

Test	Measured value
Environment	Windows Server 2022 build 20348 · AMD EPYC 9454P · 4 cores · 16 GiB · loopback
Protected 1W max stable mean	1,176.9337 RPS (range 1,163.8582–1,190.0091)
Baseline 1W max stable mean	3,442.9391 RPS (range 3,413.4320–3,472.4462)
Same concurrency 2	Baseline 3,442.9391 / Protected 1,104.7263 RPS · reduction 67.9133%
Latency c2 baseline → protected	p50 0.5468→1.7223 ms · p95 0.7238→2.2881 ms · p99 0.9382→2.6597 ms
Protected security	200,128 success / 200,128 handler · invariant failure 0 · plaintext HTTP 426 · fallback NONE
Protected 2W / 4W	NOT_MEASURED · single_worker_lease_unavailable fail-closed
Baseline scaling reference	2W 5,035.1443 RPS c8 · 4W 5,475.0515 RPS c32 · zero error

- Same business_handler, POST /secure/echo and serialization were used for Baseline and Protected.
- The 2W/4W protected values were not estimated or multiplied.
- These figures do not support a claim that PQC overhead is negligible.

Artifact authority and final scope

Artifact	SHA-256
Release 63 Product	1DCBE23F217B3C4ABC6878E5FE9942BB1F9F33D4D8267E3A12F61DEA3FA8FEC6
Final Customer Delivery ZIP	766616EC9B85D0D16C971CF5D24C3CDB14F5FD135CBBCB8CC2AC54740A232510
g269 Evidence ZIP	D2DC7D01C767D013054CEF762E3CDDCFF917D32E58B14E65D97DA0E9F3A024B7
g270 FULL AUM Evidence ZIP	92E0879EA1A256CAE23CD7A41AD09DBD8639E3AC581FF3E55B97D7BA5AB10A2E

**CUSTOMER_DELIVERY_TECHNICAL_PASS=YES · OFFICIAL_SUPPORTED_OS_6_OF_6=PASS ·
NEGATIVE_GATE=18/18 PASS**